

A) Power

$$y = x^n$$

$$\frac{dy}{dx} = \frac{d(x^n)}{dx} = n \cdot x^{(n-1)}$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial x}$$

$$\underline{x \cdot \text{grad}} = (n \cdot x^{(n-1)}) \cdot \underline{y \cdot \text{grad}}$$

B) ReLU

$$y = \max(0, x)$$

$$\frac{dy}{dx} = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$\frac{dy}{dx}, x > 0 = \frac{dx}{dx} = 1$$

$$\frac{dy}{dx}, x \leq 0, \frac{d(x-x)}{dx} = 0$$

$$\frac{\partial L}{\partial x} = \frac{\partial y}{\partial x} * \frac{\partial L}{\partial y}$$

$$x.\text{grad} = \frac{\partial y}{\partial x} * y.\text{grad}$$

$$x.\text{grad} = y.\text{grad} \quad (x > 0) \quad (\partial y / \partial x = 1)$$

$$x.\text{grad} = 0 \quad (x \leq 0, \partial y / \partial x = 0)$$

if $x > 0$:

$$x.\text{grad} = y.\text{grad}$$

else:

$$x.\text{grad} = 0$$

* $y = \frac{1}{1 + e^{-x}}$ (Sigmoid)

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{1 + e^{-x}} \right)$$

(we know that, $d(x/y) = \frac{y dx - x dy}{y^2}$)

$$\frac{d}{dx} \left(\frac{1}{1+e^{-x}} \right) = (1+e^{-x}) \overset{0}{d(1)} - (1) d(1+e^{-x})$$

$$= \frac{-d(1+e^{-x})}{(1+e^{-x})^2} = \frac{-(d(1) + d(e^{-x}))}{(1+e^{-x})^2}$$

$$= \frac{+e^{-x}}{(1+e^{-x})^2} = e^{-x} \times \left(\frac{1}{1+e^{-x}} \right)^2$$

$$= e^{-x} \times y^2$$

$$= (1+e^{-x}-1) \times y^2$$

$$\Rightarrow \left(y = \frac{1}{1+e^{-x}}, (1+e^{-x}) = \frac{1}{y} \right)$$

$$= \left(\frac{1}{y} - 1 \right) \times y^2 = \frac{(1-y)}{y} \times y^2$$

$$= (1-y) \times y$$

$$\frac{\partial L}{\partial x} = \frac{\partial y}{\partial x} * \frac{\partial L}{\partial y}$$

$$x \cdot \text{grad} = \frac{\partial y}{\partial x} * y \cdot \text{grad}$$

$$x \cdot \text{grad} = y * (1-y) * y \cdot \text{grad}$$

